

INCITE Proposal: First-principles Calculations of Self-trapped Excitons

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June 2025

1 Project details

- Broadband white light emission reported in double perovskites.
- For example, this has been shown in doped Cs₂AgInCl₆ [13] [3] [2] [15].
- Experimental efforts include using doping and different compositions to increase white light emission efficiency [13].
- Self-trapped excitons have been proposed as the cause for this broadband white light emission.
- Experimental signatures include broadband emission and stokes shift in the PL spectra.
- Even though experimental efforts exist, first-principles efforts can explain the role of phonons and doping in self-trapped exciton formation and broadband emission would be useful.
- Excited-state forces (ESF) has been used to study self-trapped excitons before as a first-principles effort [10].
- The method involved looking relaxing the structure till the excited-state force in the system is minimized.
- This method was initially developed by Beigi [9], and extended by Grande [6].
- The method has been used to look at self-trapped excitons in SiO₂ [10] and excited state forces in LiF [6].
- This method is computationally intensive, as it involves supercell calculations.
- In this project, we choose to leverage the exascale computational resources provided by Frontier and Aurora to be able to solve this problem under INCITE project.
- A first-principles understand of phonons and doping in self-trapped exciton formation can lead to engineering insights for efficient stable solid-state white lighting.
- In this project, we plan to calculate properties of self-trapped excitons from first-principles in Cs₂AgInCl₆, a double perovskite, using excited-state forces method. We will relax the structure similar to previous method on SiO₂ [10], and look at the stokes shift to compare with experimental results. Based on the displacements, the relevant phonon modes that couple strongly with the STE will also be looked at. We would also like to look at the effect of doping Na, Al, Sb, Bi on STE emission.
- Based on the estimates, we would like to request 60,000 node hours for these calculations. The calculation breakdown is given below.

2 Computational details

- The excited-state forces are calculated from exciton wavefunctions and electron-phonon coupling coefficients [6].
- Calculations are built on top of density functional theory (DFT) calculations performed using QuantumEspresso (QE) [8].
- The exciton wavefunctions are calculated by solving GW and BSE (Bethe-Salpeter equation) equations under the many-body perturbation theory (MBPT).
- The BerkeleyGW (BGW) software package is used to calculate them [7].
- The electron-phonon coupling coefficients are calculated using QE and EPW software package [14].
- The excited-state forces are calculated using the [code](#) developed by Grande [6].

The steps in the calculations are roughly:

- 1. DFT.
- 2. DFPT.
- 3. GW.
- 4. BSE.
- 5. Excited-state force (ESF) and movement in force direction.
- Repeat 1-5 till ESF is minimized.

2.1 Computational Complexity

We define the following parameters:

n_a	Number of atoms
n_ν	$3 \times$ Number of atoms
n_{uc}	Number of unit cells
n_g	Number of plane waves
n_{occ}	Number of occupied states
n_{emp}	Number of empty states (for GW)
n_{bnd}	Number of bands (occupied + empty)
n_c	Number of conduction bands (for BSE)
n_v	Number of valence bands (for BSE)

We will use $n_{c/v}$ to denote the number of conduction or valence bands for this system, as they are close. n_{nu} indicates the number of phonon modes in the system.

Based on this, the computational complexity are given by the table below. Note that since, we are considering supercells at gamma-point, no k-point considerations are needed.

Step	Equation	Time	Time (uc)
DFT	$H_{gg'}\psi_{g'}^n = E^n\psi_g^n$ [8]	$O(n_g^3)$	$O(n_{uc}^3)$
DFPT	$H_{gg'}(\partial_{s\alpha}\psi_{g'}^n) = (\partial_{s\alpha}V_g)\psi_g^n$ [1]	$O(n_g^3n_an_{occ})$	$O(n_{uc}^5)$
GW (Epsilon)	$\chi_{gg'} = \sum_{n_{emp}n_{occ}} M_{nn'g}M_{nn'g'}^*/(E_n - E_{n'})$ [7]	$O(n_g^2n_{emp}n_{occ})$	$O(n_{uc}^4)$
GW (Sigma)	$\Sigma_n^c = \sum_{gg'} M_{nn'g}M_{nn''g'}^*I_{gg'}$ [4]	$O(n_g^2n_{bnd}n_{c+v})$	$O(n_{uc}^4)$
BSE (Kernel)	$K_{cv,c'v'}^d = \sum_{gg'} M_{cc'g}W_{gg'}M_{vv'g'}^*$ [4]	$O(n_g^2n_c^2n_v^2)$	$O(n_{uc}^6)$
BSE (Absorption)	$H_{cv,c'v'}^{bse}A_{c'v'}^S = \Omega^S A_{cv}^S$ [4]	$O(n_c^3n_v^3)$	$O(n_{uc}^6)$
ESF	$F_{s\alpha} = \sum_{cc'\lambda} n_\lambda^{s\alpha} A_{cv}^{0*} A_{c'v}^0 g_{cc'\lambda} + valence\ sum$ [6]	$O(n_{c/v}^2n_a^2)$	$O(n_{uc}^4)$

where only the computationally intensive equations are listed at each step, with notational brevity. $I_{gg'}$ indicates the result of the coulomb-hole integral in GW calculations. s indicates the lattice basis

index, and α the coordinate index. η is the phonon mode. Number of plane waves scales as number of unit cells. The above is only for CPU time scaling.

As for memory usage, we parallelize as follows,

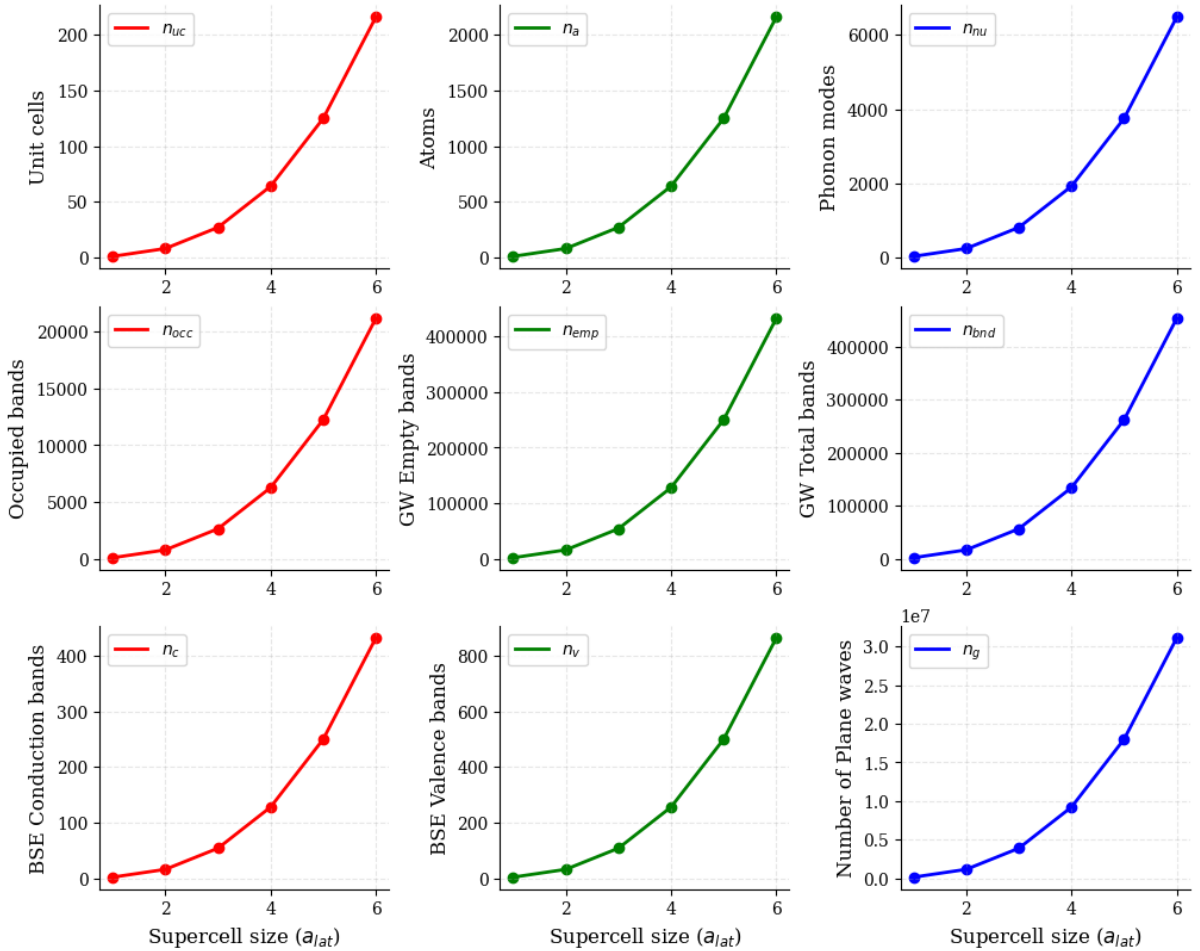
Step	Memory per node
DFT	$O(n_g n_{g,local})$
DFPT	$O(n_g n_{g,local} n_{occ,local} n_{\nu,local})$
GW	$O(n_g n_{(occ,emp),local})$
BSE (kernel)	$O(n_g n_{g,local} n_{(c,v),local}^2)$

based on available plane wave, phonon modes, and band parallelization schemes in Quantum Espresso [8] and BerkeleyGW [7] for the memory intensive parts. For ESE, no parallelization is used, as it is estimated to be comparatively fast and doesn't use much memory.

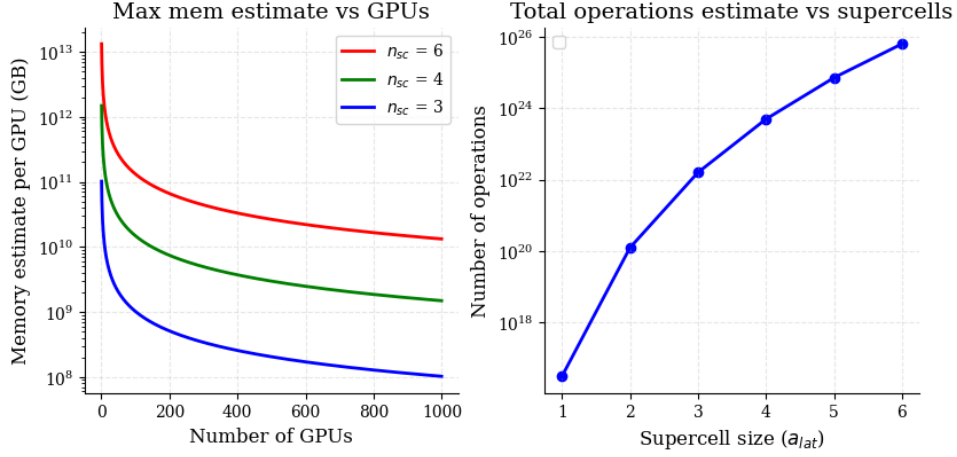
2.2 System Scaling

We assume plane waves corresponding to $0.2\text{\AA}$ as the DFT cutoff for the system. With a system lattice constant of around 10.481 and 10 atoms per unit cell [11], we can estimate the number of plane waves needed for the DFT calculations. Quantum Espresso unit cell calculations show that 92 occupied bands are necessary for SOC enabled calculations. Unit cell GW calculations require around 2000 bands for this system, and BSE calculations require around 2 conduction bands and 4 valence bands.

Based on this, we can get system scaling parameters for different supercell sizes,



We will use the DFPT parallelization to estimate memory and total number of operations. System scaling for memory sizes and times.



If the time scaling calculations are still high, machine learning methods for DFPT from learning from DFT calculations can be explored [12], and [5] for BSE calculations for supercells. It is not yet known how that scaling will proceed.

2.3 Estimates

Based on the memory and time estimates, we will use $100 \text{ nodes} \times 10 \text{ hours} = 1000 \text{ node hours}$ for computationally intensive parts and 500 node hours for others, which gives 1500 node hours for 1 single iteration.

Since this calculations is repeated to get convergence, with estimates of ~ 10 steps [10] [6], although good estimates are not available for this system, this brings the times to $15,000 \text{ node hours}$. Finally, since we want to do this for Na, Al, Sn, Bi dopings, we would need $60,000 \text{ node hours}$.

Depending on initial progress in the calculations, and available resources, we might scale down from $6 \times 6 \times 6$ supercell to $4 \times 4 \times 4$ supercells, and explore more doping fractions if there are more resources.

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